border security, and law enforcement agencies. It became a model for future miniature UAV development, influencing systems such as the RQ-20 Puma and Wasp III, which further enhanced endurance and imaging capabilities.

Figure 3.9 depicts the RQ-11 Raven (2003), a hand-launched tactical unmanned aerial vehicle developed by AeroVironment Inc. (United States) for real-time reconnaissance and surveillance. The figure highlights key components such as the forward-mounted EO/IR camera, GPS antenna, and electric motor with a rear pusher propeller, mounted on a lightweight airframe with high-mounted wings for stability and endurance. The Gimbal Control Unit beneath the nose ensures camera stabilization during flight, while the ground control station (GCS), shown to the right, provides portable command and telemetry capabilities for mission monitoring. Designed for ease of use, the Raven can be launched by hand, controlled autonomously via GPS waypoints, and recovered with minimal equipment. The figure effectively demonstrates the Raven's compact, field-ready design and its critical role in providing platoon-level tactical intelligence through live video and telemetry data transmission in modern military operations.

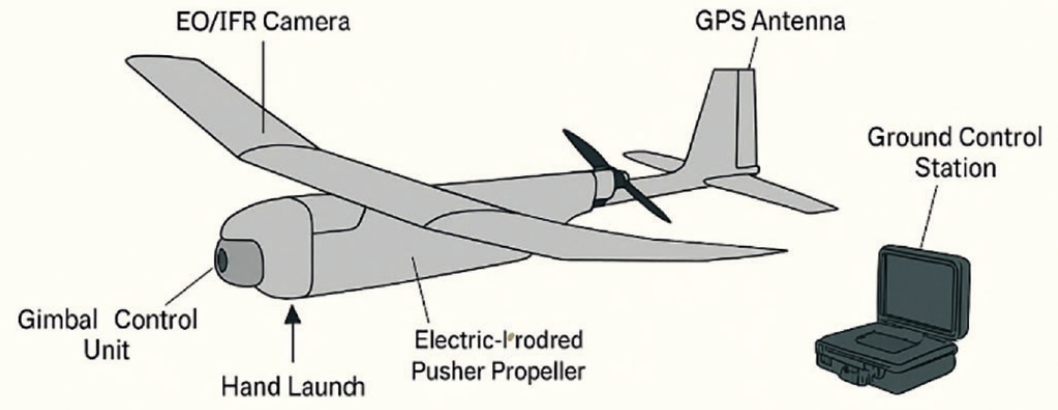


Figure 3.9 RQ-11 Raven (2003) – hand-launched tactical UAV.